INCH-POUND

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MILITARY SPECIFICATION

BATTERIES, NON-RECHARGEABLE, LITHIUM MANGANESE DIOXIDE

This Specification is approved for use by U.S. Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-rechargeable batteries of the non-reserve type composed of electrochemical cells utilizing a Lithium Manganese Dioxide system (see 6.1).

1.2 Classification.

1.2.1 Type designation. The type designation of Lithium Manganese Dioxide non-rechargeable batteries shall be in the following form (see 3.1).

BA-	5372	/U
Component (1.2.1.1)	Battery Type number (1.2.1.2)	Installation indicator (1.2.1.3)

1.2.1.1 Component. Non-rechargeable batteries are identified by the two-letter symbol "BA" followed by a hyphen.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Research Laboratory, Attn:: AMSRL-EP-RD, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.2 Battery type number. The battery type number identifies the basic design of the battery (See 3.1) and consists of a four digit number in the 5001 through 5999 series.

1.2.1.3 Installation Indicator. The installation indicator identifies the equipment the battery is used in; i.e. /PRC-25 or if "Universal", i.e. /U indicates use in various equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571 Solder, Electronic (96 to 485°C)

MILITARY

MIL-F-14072 Finishes for Ground Electronic Equipment.

MIL-F-14256 Flux, Soldering, Liquid, Paste Flux, Solder Paste and Solder Paste Flux, (For Electronic/Electrical Use)

STANDARDS

FEDERAL

FED-STD-595 Colors.

MILITARY

MIL-STD-454 Standard General Requirements for Electronic Equipment.

MIL-STD-45662 Calibration System Requirements.

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.4), samples shall be subjected to first article inspection (see 6.2) in accordance with 4.6.

3.3 Materials. Materials listed in Table I, when used, shall be in accordance with the applicable specification (see 4.4 and 6.2).

3.3.1 Metals. All metals which do not enter into the basic electro- chemical reaction of the cell shall resist, or be treated to resist corrosion in accordance with MIL-F-14072.

3.3.1.1 Dissimilar metals. When dissimilar metals which would adversely affect battery performance are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided in accordance with MIL-F-14072.

3.3.2 Insulating compounds - flow or shrinking. When tested as specified in 4.8.13 the insulating, impregnating, potting and sealing compounds shall not flow at high temperature, nor crack or draw away from the sides of a container at low temperature. Any compound used shall be non-flammable and non-toxic. Potting shall not inhibit cell vent operations (see 4.8.10). The contractor shall assure proper insulating compound curing for each battery lot.

3.4 Design and construction. Batteries shall be of the design, construction, physical dimensions, weight, and polarity as specified in 3.1.

3.4.1 Intercell connections. Intercell connections between cell blocks and between cell block and terminal shall be so insulated or positioned as to avoid contact with other conducting material. When insulated wire is soldered to terminal lugs, it shall not be bared more than 3/32 inch from the lug, nor shall it extend more than 3/32 inch beyond the lug (see 4.8.14).

3.4.2 Age of cells and batteries. The manufacturer shall certify that the age of the cells and batteries are as follows: (see 6.2)

a. Minimum age of cells, from the time of their fabrication to the time of their presentation for acceptance inspection as batteries, shall be 5 days.

b. Maximum age of cells, from the time of their fabrication to the time of their shipping date, shall be 180 days.

c. Batteries shall be submitted for acceptance inspection not more than 120 days prior to shipping date.

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3.4.3 Terminals (insulation resistance). Terminals shall be as specified on the applicable specification sheet (see 3.1) and insulation resistance shall be not less than 5 megohms when tested as specified in 4.8.8.

3.4.3.1 Metal or Plastic Covers. Metal or plastic covers shall be used in construction of cells and batteries. Construction shall be such that it shall be impossible for a battery (or any cell of such battery) to be short-circuited by coming in contact with another of the same type when placed end to end.

3.4.3.2 Flat contact terminals. Flat contact terminal dimensions shall be as specified on the applicable specification sheet (see 3.1). If a terminal is flat contact type, but no height dimension is specified, the following test shall apply. When the battery is placed flat contact down on a flat surface, only the terminal contact shall be in contact with the flat surface. Insulator or cover shall not protrude out farther than the flat contact terminal.

3.4.4 Safety features. Each cell of the battery shall contain a designed vent. This vent shall function to release over pressure in the cell and thus preclude an explosion. This feature shall operate before the battery temperature reaches 300°F (see 4.8.10).

3.4.4.1 High temperature switch. A non-resettable high temperature switch or thermal fuse shall be as specified on the applicable specification sheet (see 3.1). When tested as specified in 4.8.17, the switch shall remain closed below 180°F and open at 190 $\pm$°F.

3.4.4.1.1 High temperature switch location. The high temperature switch or thermal fuse shall be located as close as possible to the geometric center of the battery or as indicated below. For a two or three cell in-line arrangement, the high temperature or thermal fuse shall be located between any two adjacent cells. For a cluster arrangement of three or more cells, the high temperature switch or thermal fuse shall be located at the geometric center of the cluster.

3.4.4.2 Diode. Diode(s) shall be inserted in the battery as specified in the applicable specification sheet to prevent significant reverse current (see 3.1). The diode shall have the following characteristics:

Forward current, I_F = 3 amperes
Forward voltage drop, V_F = 0.65 volts maximum
Reverse current, I_R = 2 milliamperes
Reverse voltage, V_R = 40 volts

When the battery is tested to assure that the diode is functioning correctly, the charging current shall not exceed 2.0 milliamperes maximum (see 4.8.10.4).

3.4.4.3 Complete discharge device. The complete discharge device, if specified on the individual specification sheet, shall consist of a normally open switch and a resistor connected across each battery leg in series with the switch. A single switch action shall

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activate all complete discharge devices. This requirement shall not apply to any single cell battery. Switch access in battery surface shall be through a rectangular hole measuring 1/2 inch plus 0, minus 1/8 inch by 3/16 inch plus 0, minus 1/16 inch. The effective hole size shall be provided if the battery's surface is curved. The top center of the switch plunger shall be located at center of slot plus or minus 1/16 inch. The hole shall be covered by a removable label that can be made of either paper, paper products, or plastic. The top surface of switch shall be 3/16 inch plus 0, minus 3/32 inch below the outer surface of the battery where the slot is located. The switch action shall move at least 5/64 inch down and not more than 3/8 inch. The switch shall not be in the surface of the battery that contains the electrical connector. The resistor to be used in the complete discharge device shall be within the range specified and shall conform to requirement 33 of MIL-STD-454. Each resistor shall be tested to verify its value. None of the battery's fuses, diodes or thermal switches shall prevent the discharge of any cell upon activation of the complete discharge device. The contractor shall certify the value of the resistor selected and the operation of the discharge circuitry (see 4.8.10.3, 4.8.16, and 6.2).

3.4.5 Jackets. (4.8.1) The jackets shall consist of either metallic or nonmetallic material. The contents of multicell batteries shall fit snugly enough in the jackets to minimize movement of the cells. Jackets covering one or more cylindrical cells stacked end on end, and having open top and open bottom, shall be so attached to the cells as to prevent them from slipping out when held or shaken vertically.

3.4.5.1 Metallic jackets. The outside of the battery shall have a coating to protect the jacket from corrosion during or at the conclusion of any of the tests specified herein. The contents of the jacket shall be completely insulated from the metal unless otherwise specified. The jacket shall not open at any of its seams after being subjected to the test specified. The test shall be applicable to batteries weighing 5 pounds or more (see 4.8.11.1).

3.4.5.2 Non-metallic jackets. Non-metallic or plastic jackets, shall not support combustion nor emit toxic vapors when subjected to flame. The seams of the jacket shall not be open prior to or after being subjected to any test to which the battery is subjected (see 4.8.1).

3.4.5.3 Color of Jackets. The color of exposed surfaces of jackets shall match one of the following lusterless greens 34079, 34086, 34087, 34096, 34102, 34127, and 34128 per FED-STD-595 (6.2).

3.4.6 Manufacturing processes. The contractor shall maintain records on all manufacturing processes he employs for the preparation, fabrication and/or refining of cell and battery components. These records shall be available to the Government upon request. The records shall be so maintained that any cell lot and battery fabricated from the cell lot can be identified with respect to material used, batch number, components supplier, and other pertinent process data (see 6.2).

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3.4.6.1 Time order production. Battery lots shall be packaged and shipped as produced consecutively (see 6.2).

3.5 Battery voltages.

3.5.1 Battery open-circuit voltage. The open-circuit voltage shall not exceed the maximum voltage specified. (see 3.1 and 4.8.2.1).

3.5.2 Battery closed-circuit voltage. The closed-circuit voltage shall be not less than the minimum voltage specified (see 3.1 and 4.8.2.2).

3.6 Capacity. When the battery is tested for capacity as specified in 4.8.9 the time required to reach its specified minimum voltage shall be not less than the battery capacity requirement specified (3.1).

3.6.1 Initial voltage delay. When the battery is tested for capacity, the time required at the beginning of discharge for the battery units to reach minimum voltage after the load is applied shall be no more than the time specified (3.1) (see 4.8.9.1.1).

3.7 Electrolyte leakage (cell test). Each cell used in a battery shall have a leakage rate not to exceed 1.0 percent of the total electrolyte in the cell when tested in accordance with 4.8.12.

3.7.1 Cell discharge. Any cell whose capacity deviates from the average capacity of its associated cell lot by more than ± 9 percent, or which delivers less than the minimum capacity service requirement, shall be considered as failed. There shall be no leaking, venting, fire, or explosion (see 3.1 and 4.8.12.2).

3.8 Vibration. After the batteries have been tested as specified in 4.8.6 they shall meet the visual and mechanical and battery voltage requirements (see 3.4 and 3.5). There shall be no voltage fluctuations during the test.

3.9 Mechanical shock. After the batteries have been tested as specified in 4.8.5 they shall meet the visual and mechanical and battery voltage requirements (see 3.4 and 3.5). There shall be no voltage fluctuations during the test.

3.10 Drop test. Following the battery drop test, the socket shall remain within the limits specified on the individual specification sheet (see 3.1). The battery shall meet the electrical requirements specified. Following each drop, cells shall not be exposed and battery dimensions shall remain within tolerance (see 4.8.3). Defect categories 104 and 202 of table IX shall not apply to batteries that have been subjected to the drop test.

3.11 Altitude. After the batteries have been tested as specified in 4.8.7 they shall meet the visual and mechanical and battery open circuit voltage requirements (see 3.4 and 3.5.1).

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3.12 Labeling and marking. All labeling and marking shall be clear and legible throughout all the tests specified herein. Labeling and marking shall be black. Metallic and plastic jackets may have the labeling and marking engraved, or die stamped, in which case it may be the same color as the background.

3.12.1 Labels. Each battery shall have a label as specified in 3.1. If there is insufficient space to show all required information on one face of the battery, it shall be continued on another face. There shall be no information on the label other than the following:

BATTERY, NON-RECHARGEABLE, LITHIUM MANGANESE DIOXIDE
Type Designation
(Contract Number) --
(Code) -- (See Note and 3.12.1.1)
Serial Number (See 3.12.1.2)
Manufacturer's name
(Trade name may also be-used)
Manufacturer's plant
DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, OR MUTILATE THIS BATTERY OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS

EXAMPLE:

BATTERY, NON-RECHARGEABLE, LITHIUM MnO2
BA-5270/U
DAAB05-83-C-1234
0383B

Serial number 1-2-3
James E. Doe Company

JODOCO
Batteryville, NJ

DO NOT CHARGE, SHORT CIRCUIT, INCINERATE OR MUTILATE THIS BATTERY OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS

Note: The code may be placed on the bottom of single-cell batteries.

3.12.1.1 Code. The code shown shall indicate the month, year and week of manufacture of the battery by means of a four-digit number followed by a single letter. The first two digits shall indicate the number of the month. The last two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0". "A" shall be used for the first week of the month, "B" for the second week of the month, etc. Sunday shall be considered the first day of a week.

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EXAMPLES:

A battery manufactured during the second week of March 1983 will bear the code "0383B".

A battery manufactured during the third week of November 1983 will bear the code "1183C".

3.12.1.2 Serial number. Each battery produced and furnished under a particular contract shall be given a specific serial number. These serial numbers shall be assigned consecutively beginning with one and continuing therefrom for the entire contract lot of batteries. Serialization of batteries shall be accomplished immediately upon completion of fabrication and prior to sampling for inspection. Serial numbers of batteries are not required on shipping containers or packaging.

3.12.2 Terminal marking. On batteries having socket-type terminals, all markings such as polarity, voltage, and the voltage unit of battery (A, B, C, etc.) shall appear on the face of the battery bearing the socket. On other type terminals, the terminal markings may appear on the top or the side of the battery, or both. Markings shall indicate clearly the terminals to which they refer.

3.13 Cell water content. Water content of the electrolyte solution in a cell shall not exceed 200 parts per million by weight when tested by the Karl Fischer electrolytes titration method. The contractor shall certify that this limit is not exceeded throughout production (see 6.2 and 4.8.15).

3.14 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, interchangeability, or appearance in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's

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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required examinations and tests. Unless otherwise specified herein, all examinations and tests shall be performed under ambient temperature, humidity, and atmospheric pressure conditions. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.2.1 Instrument accuracy.

4.1.2.1.1 Voltmeters and ammeters. All voltmeters and ammeters used in testing the batteries shall be accurate within 1 percent of the full scale value. The voltmeter and ammeter ranges shall be such that all readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be not less than 10,000 ohms per volt.

4.1.2.1.2 Resistor tolerance. During all tests involving discharge through a resistance, such resistance shall be accurate within the following percentages:

	Percent
Up to and including 25,000 ohms	± 0.5
From above 25,000 ohms to and including 1 megohm	± 1.0
Above 1 megohm	± 5.0

In determining the resistance used as a test load, the resistance of all continuously operating voltmeters shall be considered as part of the specified load.

4.1.2.1.3 Power supplies. Power supplies used for discharges specified herein shall be accurate within ± 1 percent.

4.1.2.1.4 Timing. Timing equipment shall be accurate within 0.5 percent.

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4.2 Classification of inspection. The examination and testing of batteries shall be classified as follows:

a. Materials inspection (see 4.4).

b. In-process inspection (see 4.5).

c. First article inspection (see 4.6).

d. Quality conformance inspection. (see 4.7)

4.3 Inspection conditions. Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80±20°F.

4.4 Materials inspection. Materials inspection shall consist of verifications by certification from the source that the materials used in fabricating the cells and the batteries are in accordance with applicable requirements prior to such fabrication. In the absence of certification from the source, a certificate of analysis or certified inspection data shall be required as proof of conformance to applicable requirements. (see Table I).

TABLE I - Materials inspection.

Material	Requirement Paragraph	Applicable Specification
Metals and Finishes Solder (SN40, SN60, SN63) Solder Flux (type RMA)	3.3.1	MIL-F-14072 QQ-S-571 MIL-F-14256

4.5 In-process Inspection. Each inspection lot shall be inspected in accordance with Table II including the cells made for the first article inspection.

4.5.1 Cell lot. A cell lot shall be defined as the quantity of cells manufactured during a period of time of up to one month, or the cells produced on one date of any one type and produced at any one place of manufacture by any one production method.

4.5.2 Sampling plan. A cell sample lot shall consist of 30 cells.

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TABLE II - In-process inspection.

Inspection	Requirement Paragraph	Test Method Paragraph	Cell	Battery Component	Sampling Plan
Electrolyte leakage	3.7	4.8.12	X		4.8.12.1
Cell water content	3.13	4.8.15	X		Certification only
Insulating compounds	3.3.2	4.8.13.		X	Once per battery lot
Intercell connections	3.4.1	4.8.14		X	4.8.14.1
High temperature switch	3.4.4.1	4.8.17		X	4.8.17.1
Complete discharge device	3.4.4.3	4.8.16		X	100 percent of resistors

4.6 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.6.1 Sampling plan. The number of batteries constituting an inspection lot shall be in accordance with Table III.

4.6.2 Inspection routine. First article inspection shall consist of all the examinations and tests in accordance with Table III. One sample battery, untested is to remain at the contractor's plant and is to be available as a standard for comparative purposes.

4.6.2.1 First article data. The first article test plan and test report(s) shall be as specified in the contract or purchase order (see 6.2)

4.6.2.2 Failure. If one or more sample batteries fail to meet any of the first article requirements and tests, the contractor shall immediately make the remedial changes. The contractor at no additional cost to the government shall be required to submit additional first article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the first article inspection reports.

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TABLE III - First article inspection.

Group	No of Batteries	Examination and Test	Required Paragraph	Method of Test Paragraph
I		Visual & mechanical examination	3.4, 3.12	4.8.1
		Battery voltage	3.5	4.8.2
		Dimension & weight	3.4	4.8.4
	20	Mechanical Shock	3.9	4.8.5
	20	Vibration	3.8	4.8.6
		Altitude	3.11	4.8.7
		Insulation resistance terminals	3.4.3	4.8.8
		Drop test	3.10	4.8.3
		Capacity	3.6	4.8.9
IA	10	"I" Test @ 70°F	3.6	4.8.9.1.2
IB	5	"L" Test @-20°F	3.6	4.8.9.1.3
		Complete discharge device	3.4.4.3	4.8.10.3
IC	5	"H" Test @ 130°F	3.6	4.8.9.1.4
		Complete discharge device	3.4.4.3	4.8.10.3
II		Visual-mechanical	3.4, 3.12	4.8.1
	25	Battery voltage	3.5	4.8.2
		Capacity	3.6	4.8.9
IIA	10	"HT" Test @ 130°F after 4 weeks storage @ 130°F	3.6	4.8.9.1.6
	25			
IIB	10	"LT" Test @-20°F after 4 weeks storage @ 130°F	3.6	4.8.9.1.5
IIC	5	"ILR" Test @ 80°F	3.6	4.8.9.1.9
III	10	Battery Voltage	3.5	4.8.2
IIIA	10	Safety features	3.4.4	4.8.10
IV	1	Untested Sample	----	4.6.2

4.7 Quality conformance inspection.

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4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A, B, and C inspection. Test equipment for government verification inspection shall be made available by the contractor, if required.

4.7.1.1 Inspection Battery Lot. A lot shall be defined as the quantity of batteries of any one type produced at any one place of manufacture on any one contract during one month or less period, submitted at one time to quality conformance inspection.

4.7.1.1.1 Shipment lot. The shipment lot is the quantity of batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more shipment lots, under the terms of any one contract.

4.7.1.1.2 Contract lot. The contract lot is the total of all batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more shipment lots, under the terms of any one contract.

4.7.1.2 Group A inspection. Each battery on contract or purchase order shall be 100 percent inspected for conformance to the inspections in the order specified in Table IV. Discrete lots shall be formed from batteries that pass this inspection. Recording of absolute measurements of electrical parameters is required only for failures.

TABLE IV - GROUP A INSPECTION.

Examination and Test	Requirement Paragraph	Method of Test Para.
Visual & mechanical	3.4, 3.12 and table IX	4.8.1
Battery Voltage	3.5	4.8.2
Diode test	3.4.4.2	4.8.10.4

4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in Table V in the order shown. Sample size shall be twenty-five (25) randomly selected samples, twenty (20) for subgroup I and five (5) for subgroup II. Group B inspection shall be performed on sample units from lots which have been subjected to and passed group A inspection.

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TABLE V - Group B inspection.

Examination and test	Requirement paragraph	Method or test Paragraph	Criteria Accept Number	Criteria Reject Number
Subgroup I				
Dimensions & Weight	3.4	4.8.1	0	1
Insulation resistance	3.4.3	4.8.8	0	1
Safety feature	3.4.4	4.8.10	0	1
Subgroup II				
Electrolyte leakage	3.7	4.8.12	0	1

4.7.1.4 Group C inspection. Group C inspection shall consist of (a)HT capacity test and (b) LT capacity test. Each capacity test shall be preceded by both the mechanical shock and vibration tests described in paragraphs 4.8.5 and 4.8.6 respectively.

4.7.1.4.1 Sampling plan. Samples shall be selected in accordance with Table VI and shall represent a shipment lot. Fifty percent (50%) shall be used for the HT capacity test and fifty percent (50%) for the LT capacity test.

4.7.1.4.2 Group C failures. Accept/reject criteria of Table VI shall be used for failures (a), (b) and (c). No failures shall be allowed for (d).

a. Insufficient service.

b. Excessive initial voltage delay.

c. Battery exceeds dimensional tolerances after discharge.

d. Battery vents, leaks, burns or ruptures.

TABLE VI - Group C inspection.

Inspection Lot Size	Sample Size	Acceptance No. 4.7.1.4.2 a, b, c	Acceptance No. 4.7.1.4.2 d
500 or less	12	1	0
501-1,200	20	2	0
1,201-10,000	32	3	0
10,001-35,000	50	5	0
35,001 or more	80	7	0

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4.7.1.5 Group D inspection. Group D inspection shall be performed at the Government inspection facility on sample batteries in accordance with Table VII. Shipment of the lot represented by the sample batteries shall not be held up pending the results of group D inspection.

TABLE VII - Group D inspection.

Storage and Capacity Test	Requirement Paragraph	Method of Test Paragraph
Group I (Adjustment Purposes see 4.7.1.5.3)		
Capacity T	3.6	4.8.9.1.7
Capacity D	3.6	4.8.9.1.8
Capacity ILR	3.6	4.8.9.1.9

4.7.1.5.1 Sampling plan. A sample of n_S batteries shall be selected at random from production for each shipment lot in amounts determined from the following formula, and rounded off in case of fractions to an adjacent integer (up or down for each shipment lot), so that exactly n batteries have been assigned to both T and D capacity tests when the sample for the final shipment of the contract lot has been drawn.

$$n_S = a + (N_S/N) \times 2n$$

Where:

a= 10 batteries for ILR capacity testing
n_S= total number of sample batteries to be taken from each shipment lot
N_S= number of batteries in the shipment lot
N= number of batteries in the contract lot (see table VIII)
n= number of batteries to be taken from the contract lot for each of two capacity tests, T and D, in accordance with Table VIII (total number of T and D samples selected is 2n)

4.7.1.5.1.1 Allocation of sample batteries for Group D inspection. The number of batteries, n_S, selected from a shipment lot, shall be assigned at random at the Government test facility, so that 10 batteries shall be submitted for the ILR capacity test and the remainder (n_S-10) split approximately in half for the T and D tests. Such assignment shall result in the allocation of exactly n batteries to both T and D capacity tests after the final shipment on the contract lot is made. If necessary, the sample size n_S taken from the last shipment lot of the contract lot shall be adjusted so that this result is achieved.

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TABLE VIII - Sample size and Acceptance Number for Both T and D Tests.

Contract Lot Size "N"	Sample Size "n" for T and D test from contract lot	1/ Acceptance No for T and D tests
0 to 110	5	2/ --
111 to 500	15	3
501 to 800	25	5
801 to 1,300	35	7
1,301 to 3,200	50	9
3,201 to 8,000	75	13
8,001 to 22,000	110	18
22,001 to 110,000	150	24
over 110,000	225	34

1/ When the number of capacity values falling below the minimum requirements specified (see 3.1) for a given test is equal to or less than the associated acceptance number, the contract lot from which the samples were drawn have met the requirements of that test.

2/ Determination of compliance specified in 4.7.1.5.2 shall not apply to lot sizes of less than 111.

4.7.1.5.2 Compliance. The entire contract lot shall be considered as complying when the T and D tests show compliance as specified in Table VIII, and the number of ILR failures for the contract lot does not exceed twenty (20) percent of the number of samples submitted to ILR testing.

4.7.1.5.3 Noncompliance. If the capacity test results do not show compliance with the requirements as defined in 4.7.1.5.2, the entire inspection battery lot shall be considered as not complying with requirements of this specification and an adjustment (6.2) can be made if it is determined to be in the Government's best interest by the contracting officer.

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TABLE IX - Classification of visual and mechanical examination defects.

Categories	Defects
001	Improper assembly causing parts to be inoperative or unsafe in service.
002	Electrolyte leaking caused by missing or defective sealing or closing.
003	Welded or soldered connections improperly made so as to adversely affect battery performance or create burrs that may short battery.
102	Cell aging requirement not met.
103	Contact surfaces obstructed by insulation material so that electrical use is affected.
104	Torn non-metallic jackets - any tear or rip with dimension greater than 1/2 inch.
105	Improper jacket closure.
106	Insulating parts or materials missing, damaged, or improperly located so as to affect electrical performance.
107	Location, polarity and marking of terminals not as specified.
108	Labeling and marking wrong, missing or illegible so that utilization is affected.
109	External and internal threads missing, wrong size or so damaged to prevent proper use.
201	Improper assembly which could reduce efficiency of operation but not render battery inoperative or unsafe in service.
202	Deformed or damaged parts which do not adversely affect electrical performance.
203	Burns or imperfections which do not interfere with proper use in operation, assembly or disassembly, or cause unsafe condition in service.
204	Improper marking which doesn't hamper utilization or identification of the battery.

4.8 Test Methods and examinations.

4.8.1 Visual and mechanical examination. Batteries and cells shall be examined to determine compliance with all applicable requirements and characteristics as specified herein (see 3.4, 3.12, and Table IX.).

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4.8.2 Battery Voltage.

4.8.2.1 Open-circuit voltage. A direct current voltmeter of appropriate range and sensitivity shall be used to measure the open- circuit voltage (see 3.5.1).

4.8.2.2 Closed-circuit voltage. A direct current voltmeter of proper range and sensitivity shall be used to measure the closed circuit voltage utilizing the load specified (see 3.5.2).

4.8.3 Drop test. Each battery shall be dropped once for each temperature, from a height of 30±2 inches onto a hard surface consisting of concrete. The side of the battery perpendicular to and nearest the socket shall be parallel to the concrete surface upon release and shall make contact with it. The drop test shall be performed at 160°F and -20°F. The batteries shall be stabilized a minimum of 4 hours at each test temperature and dropped within 10 minutes after removal from the temperature chamber (see 3.10).

4.8.4 Dimensions and weight. Batteries shall be examined by gauging or measuring and by weighing to determine conformance (see 3.4).

4.8.4.1 Dimensions. All dimensions shall include any coating which may be used, and shall remain within the specified tolerances throughout the required tests. Both minimum and maximum dimensions shall be determined. When box gauges are used, batteries, loaded with the following weights shall pass freely through the applicable gauge openings:

a. Batteries weighing less than 5 pounds - loading weight of 5 pounds.

b. Batteries weighing 5 pounds or more - loading weight equal to the weight of the battery.

The dimensions of the box gauge shall be the specified maximum outside dimensions of the battery.

4.8.5 Mechanical shock. Each battery shall be secured to the testing machine by means of a rigid mount which shall support all mounting surfaces of the battery. Each battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be applied in a direction normal to a face of the battery. The faces of the battery are identified by their position in relation to the front face (the face which bears the label or marking). For each shock, the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (G). The peak acceleration shall be between 125 and 175G, exact value shall be recorded (see 3.9).

4.8.6 Vibration. Each battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the batteries are clamped when in use (see 3.1). A simple harmonic motion shall be applied having an

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amplitude of 0.03 inch (0.06 inch total maximum excursion). The frequency shall be varied at the rate of 1 hertz per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95±5 minutes for each mounting position, (direction of vibration) of the battery. The batteries shall be vibrated in three equal periods in mutually perpendicular directions, one of which shall be perpendicular to the terminal face of the battery. Open circuit voltage shall be observed for 30 seconds during the last quarter of each of the three vibration periods (see 3.8).

4.8.7 Altitude. Batteries shall be placed in an altitude chamber, in which the pressure is maintained at a value corresponding to an altitude of 50,000 feet and the temperature is kept at 75±5°F for a period of six (6) hours. Open circuit voltage only shall be observed after the test (see 3.11).

4.8.8 Insulation resistance (terminals). Insulation resistance test shall be performed, except as otherwise specified (see 3.1). Batteries shall be stored for a period of 48 hours at +70±5°F and a relative humidity of 50±15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct-current potential of 500±20 volts between any two terminals not electrically connected and between all ungrounded terminals and the container of the battery. The insulation resistance of batteries having a non-metallic container shall be measured by the use of a 1 inch-square copper plate making physical contact with the container. The plate shall be placed with the broad surface against any area of any surface of the jacket other than that on which the battery terminals are located (see 3.4.3).

4.8.9 Capacity (see 3.6).

4.8.9.1 Capacity Tests. Sample batteries selected for capacity tests specified in the individual specification sheet (3.1) shall be stored and discharged in air as applicable in accordance with 4.8.9.1.1 through 4.8.9.2.1. All batteries shall be discharged to 0 volts, except those subjected to the ILR test. The time required to reach the specified minimum voltage shall be used to determine the battery capacity.

4.8.9.1.1 Initial voltage delay. At the start of the capacity discharge test, batteries shall be monitored with an oscillographic device or equivalent to determine the time in milliseconds required for battery units to reach minimum voltage after the specified loads are applied as specified in the individual specifications sheets. No loads shall be applied at any time during storage or prior to the discharge test following such storage (see 3.6.1).

4.8.9.1.2 Capacity test I. Discharge at +70±5°F without previous storage.

4.8.9.1.3 Capacity test L. Discharge at -20±3°F after storage at-20±3°F for a minimum of sixteen (16) hours.

4.8.9.1.4 Capacity test H. Discharge at +130±3°F after storage at +130±3°F for a minimum of sixteen (16) hours.

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4.8.9.1.5 Capacity test LT. Discharge at -20±3°F after four (4) weeks storage at 130°F followed by a minimum of sixteen (16) hours at -20±3°F. No load shall be applied during the storage at 130°F and -20°F or prior to discharge test.

4.8.9.1.6 Capacity test HT. Discharge at +130±3°F after four (4) weeks storage at 130°F and a minimum of sixteen (16) hours at +130±3°F.

4.8.9.1.7 Capacity test T. Discharge at +70±5°F after thirteen (13) weeks storage at +130°F.

4.8.9.1.8 Capacity test D. Discharge at +70±5°F after fifty-two (52) weeks storage at +70°F.

4.8.9.1.9 Capacity test ILR. Discharge at +80±20°F to the specified minimum voltage without previous storage.

4.8.9.2 Storage conditions. The storage conditions specified in Table X shall prevail during storage periods specified. Normal conditions shall be maintained. No loads shall be applied at any time during storage or prior to the discharge test following storage. Deviations from normal conditions are permitted provided that (1) the extreme conditions specified in Table X do not exist for more than ten percent (cumulative of the specified storage periods) and (2) that at no time are the extreme conditions exceeded. Batteries shall be oriented in storage to have at least 50 percent of the cell seals at the bottom of the batteries (see 3.1).

Table X - Storage test conditions.

Kinds of storage	Normal conditions temperature	Extreme Conditions temperature
T, HT, and LT storage	+130 ±3°F	108°F thru 127°F and 133°F thru 135°F
D storage	+70±5°F	60°F thru 65°F and 75°F thru 80°F
Temperature cycling	+160 +3°F -7°F	140°F thru 153°F and 163°F thru 165°F

4.8.9.2.1 Discharge. Following stabilization the batteries shall be discharged at the ambient discharge conditions as specified. There shall be at least a two (2) inch separation between all batteries that are being discharged. All batteries subjected to capacity testing shall

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be discharged as specified in individual specification sheets. A battery shall be considered to have failed a discharge test when any one of the following conditions occurs:

a. Insufficient service.

b. Excessive initial voltage delay.

c. Battery exceeds dimensional tolerances after discharge.

d. Battery vents, leaks, burns or ruptures.

4.8.10 Safety feature. Fifty percent of batteries set aside for safety feature test shall be tested according to paragraph 4.8.10.1. The remaining batteries shall be tested according to paragraph 4.8.10.2.

4.8.10.1 Short circuit test. Each battery shall be subjected to a direct short, i.e., less than .05 ohms, for a period of time not less than 24 hours or until the battery temperature returns to ambient. At the end of this test, the battery shall be checked that no cell vented. Batteries containing cells which leaked, exploded, burned, or vented are failures. Distortion of labeling or dimensions is permitted.

4.8.10.2 Vent test. All batteries shall be placed at +180°F for two hours. At the end of the two hours the battery shall be checked to see that no cell has vented. If any cell has vented, the battery has failed this test. If no cell has vented, the ambient temperature shall be raised to a maximum of 300°F. Batteries shall be maintained at an ambient of +300°F for two hours. All cells shall have vented at the designated vent after the two hours at 300°F. If the cell has a crimp seal construction, (i.e.: rubber, plastic), then venting is also allowed at the crimp seal. If the cell is hermetically sealed, then venting is only permitted at the designed vent. Batteries containing cells which exploded, burned or did not vent are failures (see 3.4.4). Distortion of labeling or dimensions is permitted.

4.8.10.3 Complete discharge device. Location and operability shall be verified. Ability to discharge an end voltage discharged battery after activation of complete discharge device shall be certified and verified during first article inspection (see 3.4.4.3 and 6.2). The device shall be activated on the five batteries submitted to the Group 1B test and also the five batteries submitted to the Group 1C test. The device shall apply the specified load to the battery or its section when the switch is activated (see 3.4.4.3 and 6.2).

4.8.10.4 Diode breakdown test. A DC power supply capable of delivering at least 2.50 ma. shall be used. The voltage to be used shall be 40 (+0, -1) volts, plus the voltage obtained by multiplying the number of cells in series by 3.05. It shall be electrically connected with low impedance contacts to the connector terminals of series-connected cell strings of the battery to force reverse current flow (charging) through the individual cell string (i.e., positive to positive and negative to negative). This voltage shall be applied for a minimum of 1.0 second. This test shall be conducted immediately following the battery

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closed-circuit voltage test. The amount of current flowing shall not exceed the amount specified (see 3.4.4.2).

4.8.11 Jackets.

4.8.11.1 Metallic jackets. Metallic jacketed batteries weighing five pounds or more shall be loaded by applying weights totaling 100 pounds evenly distributed over the top of the battery and shall remain so loaded at least one minute. The condition of the jacket shall be observed.

4.8.12 Electrolyte leakage test. At no time shall there be potting substances or cell jackets applied to the cells designed for this test. The cells shall be numbered and weighed to the nearest milligram. The weight shall be recorded and the amount of electrolyte in each cell known. Fifty percent of these cells shall be stored for four weeks at 130°F. The remaining cells shall be temperature cycled. They shall be cycled for 12 hours at +160°F (71°C) and 10 hours at -60°F (-51°C) and 1 hour in between to change temperatures. This 24 hour cycle shall be repeated 7 times for a total time of 7 days. At the end of each test, the cells shall be removed and allowed to cool at room temperature for at least two hours. Each cell shall then be weighed to the nearest milligram. The weight loss shall be recorded. The weight loss shall not exceed 1.0 percent of the total electrolyte weight. The electrolyte salt weight shall not be used in this computation.

4.8.12.1 Sample size. The number of cells examined per inspection battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.8.12.2 Cell discharge test. Following the electrolyte leakage test, the same filled cells used in the test shall be discharged at 70±5°F through the load and to end voltage in accordance with the individual specification sheet. There shall be at least a two (2) inch separation between all cells that are being discharged. Following capacity determination, the discharge shall continue to zero volts (see 3.1 and 3.7.1)

a. If the filled cell samples meet the requirements of 3.7.1, the cell lot may be used for fabrication of batteries.

b. If there are two (2) or more failures (delay or capacity), during the capacity testing of the cell sample, the cell lot shall be rejected. If only one (1) failure occurs, the cause of the failure shall be determined and those cells with the defect shall be culled from the lot represented. If the cause cannot be determined, the lot shall be rejected. No venting, leaking, fire or explosion of cells shall be allowed.

4.8.13 Flow or shrinking (insulating compounds) (see 3.3.2). Compounds shall be placed in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within 1/4 inch of the top. The temperature of the compound within the container shall be raised to +200±5°F and the container shall be held in an inverted position for 24 hours. Then the temperature of the compound shall be lowered to -40±5°F ambient for eight hours

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minimum. This test shall be performed where required either during First article inspection or during quality conformance inspection for the first battery lot.

4.8.13.1 Sample Size. The number of cells examined per inspection battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.8.13.2 Failure Rate. The failure rate shall be the same as for the "T" capacity test in Table VIII.

4.8.14 Intercell connections. Intercell connections shall be visually examined for conformance to 3.4.1.

4.8.14.1 Sample size. Sample size shall be twenty each for each shipment lot. No failures allowed.

4.8.15 Cell water content. Cell water content shall be certified for first article inspection in the first article test report and for each inspection battery lot in the quality conformance test report. (see 3.13 and 6.2).

4.8.16 Complete discharge device resistor. Each battery shall have the value of the resistor to be used in the complete discharge device measured by applying a voltage and reading the resulting current prior to installation in the battery (see 3.4.4.3)

4.8.17 High temperature switch test. This test shall be performed on switches or thermal fuses prior to installation in batteries. Test sample shall be placed in a temperature chamber at $+175 \pm 5°F$ for a minimum of two hours. each sample shall then be checked to verify that the switch or thermal fuse is closed. The temperature shall be raised to $+190 \pm 5°F$. after 45 minutes, each sample shall be checked to verify that the switch or thermal fuse is open (see 3.4.4.1).

4.8.17.1 Sample size. The number of high temperature switches or thermal fuses tested per inspection battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.8.17.2 Failure rate. the failure rate shall be the same as for the "T" capacity test in table IX.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

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6. NOTES

6.1 Intended use. The primary batteries included are of the non-reserve type composed of electrochemical cells utilizing a lithium manganese dioxide system. The lithium cell batteries are capable of storage and use under wide temperature ranges.

6.1.1 Indirect shipments. The packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipment to the Government.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable specification sheet.

c. Complete type designation (see 1.2.1).

d. Requirement for first article test plan and test report (see 4.6.2.1).

e. Requirement for maintaining manufacturing processes (see 3.4.6).

f. Requirement for certification of materials and processes (see 3.3).

h. Requirement for certification of age of cells (see 3.4.2).

i. Requirement for certification of age of batteries (see 3.4.2)

j. Requirement for contractor to state color of jackets (see 3.4.5.3).

k. Requirement for certification of operation of complete discharge device for production lots (see 3.4.4.3 and 4.8.10.3).

l. Adjustment procedures for Group D inspection for Capacity (see 4.7.1.5.3).

m. Requirement for certification of cell water content (see 3.13).

6.3 Definitions.

6.3.1 Non-flammable and non-toxic materials. Non-flammable and non-toxic materials are those materials which will not support combustion, produce smoke, or be capable of emitting toxic fumes when subjected to the environmental conditions specified for the battery.

6.4 First article. When first article inspection is required (see 3.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be a

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preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.6. The contracting officer should also include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 International standardization agreement. Provision of this specification are the subject of international standardization agreements. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.6 Subject term (key word) listing.

Cells
Complete Discharge Device
Voltage, Open Circuit

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians
Army - ER

Preparing Activity
Army - ER

(Project 6135-A268)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-49458A(ER)	940721

3. DOCUMENT TITLE
Batteries, Non-Rechargeables, Lithium Manganese Dioxide

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Army Research Laboratory	(908) 544-3148	995-3148
c. ADDRESS (Include Zip Code) ATTN: AMSRL-EP-RD Ft. Monmouth, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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